

Victor Li

NYC Metropolitan Area & Ann Arbor, MI | victli@umich.edu | +1 718-640-4007 | [LinkedIn](#) | [Portfolio](#)

EDUCATION

University of Michigan Engineering | GPA: 3.8 | B.S.E. BME | Expected 2028 | Jim and Denise Wartinbee Scholarship Recipient
Bronx High School of Science, Bronx, NY | GPA: 4.0 | SAT: 1560

Relevant Courses: Intro Programming, Engineering 100, Calculus 1-4, Biology, Chemistry, Physics 1 & 2, Statistics

TECHNICAL & ANALYTICAL SKILLS

Data Analysis: Python (pandas, NumPy), R (tidyverse), Java (OOP), SQL fundamentals, MATLAB, Predictive Modeling/ML

Spreadsheet & Reporting: Excel (pivot tables, ApachePOI), R Studio, Jupyter Notebook, Microsoft Office

Visualization & Tools: R Shiny, Biorender, Git/GitHub, HTML/CSS, Adobe Photoshop, AlphaFold Multimer, VMD, CAD

Wet Lab & Molecular Biology: Protocol execution, micropipetting, solution prep, PCR, gel/capillary electrophoresis, BLAST

WORK EXPERIENCE

BME & ML Lab Researcher | Chandrasekaran Lab | University of Michigan, Ann Arbor **August 2025–Present**

- Screened 100,000+ drug combinations for synergy and built the workflow into a reusable MATLAB for future predictions
- Integrated M2D2-generated drug-protein interaction features w/ TACTIC's transfer-learning pipeline to predict antimicrobial interactions across species, replacing traditional chemogenomic inputs with structure & sequence-derived ML features.
- Predicted top-performing drug combinations and translated outputs into ranked shortlists for the physical testing stage
- Delivered one-page briefs & Excel deliverables; presented weekly updates in lab meetings to support faster, clearer decision

NSF–MPS Research Intern | UCLA & NSF | Remote **Jun 2023–Mar 2025**

- Analyzed and cleaned 2.1M+ continuous glucose monitoring readings in R to support an accepted diabetes paper
- Built 30+ ggplot2 longitudinal plots w/ multi-year CGM glucose trajectories & recurring patterns across ~2.8K patients
- Developed an R Shiny dashboard consolidating CGM trend outputs and patient-level characteristics into an interactive interface for collaborator review and interpretation.
- Used GitHub to manage version control, resolve merge conflicts, and coordinate code contributions with the research team

EMHS Research Intern | Albert Einstein College of Medicine | Bronx, NY **Jun 2024–Dec 2024**

- Authored a diabetes manuscript on the inceptor–insulin receptor interaction, applying AlphaFold-Multimer, virtual screening, molecular docking, and binding-energy analysis to study mechanisms of insulin resistance.
- Developed 3 Java parsers with Apache POI to convert raw PDB outputs into structured Excel tables and identify binding sites
- Ran a screening pipeline to evaluate 6507 candidate ligands and characterize mutation-induced changes in complex stability

Molecular Biology Intern | Stanley Manne Research Institute | Bronx, NY **Dec 2024–Jun 2025**

- Executed protocol-driven bench workflows (micropipetting, PCR); tracked conditions and documented in ELN
- Ran capillary & gel electrophoresis for product QC; aggregated results (Excel) & summarized trends for review
- Used BLAST for sequence validation/primer checks; collaborated on troubleshooting whilst following lab safety

LEADERSHIP EXPERIENCE

PeriOperative Project Team Firmware Engineer | MHEAL | Ann Arbor **Aug 2025–Present**

- Developed Raspberry Pi firmware (Python + GPIO) for a relay-driven heated OR mattress, adding touchscreen UI, alerts, data logging, and safety interlocks to improve usability in low-resource operating rooms
- Validated relay electrical performance for system integration using Arduino test code and hands-on wiring/harness prep
- Ran bench tests on temperature sensors and heating profiles and analyzed results in Excel to recommend setpoints, thresholds, and safety limits; maintained GitHub documentation to help new members onboard faster

Director of Content | Youth Medical Association | NYC **Sep 2022–Dec 2025**

- Led and analyzed outreach for 30K+ monthly viewers and gave biweekly health presentations to 100+ members
- Coordinated 30+ educational tours; managed logistics across 30 global chapters and 4 teams with MS Office
- Raised \$2.5K+ for cancer research; helped form NYC Mayor's Youth Council to expand health education
- Earned Gold Level Presidential Volunteer Service Award (509 hours) for collaborative service across global chapters

PUBLICATIONS

1. Okuno T, Sort L, Zhang B, Zhou K, Kitchen M, Li V, et al. "Temporal Glycemic Patterns in Type 1 and Type 2 Diabetes: Insights From Extended Continuous Glucose Monitoring." *Journal of Diabetes Science and Technology*. Accepted.
<https://doi.org/10.1177/19322968251341264>
2. Structure-Based Computational Analysis of Interactions between Insulin Receptor and Insulin Inhibitory Receptor Victor Li, Yinghao Wu bioRxiv 2024.09.06.611694; doi: <https://doi.org/10.1101/2024.09.06.611694>